

IoT-Based Home Automation Using Digital Logic

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Internship

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Project Overview

This project demonstrates a smart home automation system using digital logic concepts. The system interfaces sensors and actuators and allows appliance control through Wi-Fi commands. The design focuses on automation and power optimization.

Objectives

- Interface sensors and actuators using digital circuits.
- Implement appliance control using Verilog HDL.
- Support remote control through Wi-Fi.
- Demonstrate power-saving techniques.

Features

- Motion Sensor Based Lighting Control
- Temperature Controlled Fan Operation
- Wi-Fi Remote Appliance Control
- Power Optimization
- FPGA Compatible Design

Components

Sensors

- Motion Sensor (PIR)
- Temperature Sensor

Actuators

- LED Light
- Fan

Communication

- Wi-Fi Module (ESP8266 Concept)

Files Included

- home_automation.v
- home_automation_tb.v
- README.md
- Project_Report.pdf

Working Principle

Light Control

- Motion detected → Light ON
- Wi-Fi command received → Light ON
- No motion and no command → Light OFF

Fan Control

- Temperature detected → Fan ON
- Normal temperature → Fan OFF

Applications

- Smart Homes
- Smart Buildings
- Energy Management Systems
- IoT Automation

Conclusion

The project successfully demonstrates digital logic-based home automation with sensor monitoring, remote control capability, and energy-efficient operation.